

Dynamic Joint Learner: Analysis on Sensitivity to F

Configuration of the Simulation

We conduct a sensitivity analysis to assess the effect of the number of NN-GP layers, F , used to fit the Dynamic Joint Learner. The data are generated under three configurations of the true latent dimension R^* and number of NN-GP layers F^* : $(R^*, F^*) \in \{(4, 1), (3, 2), (2, 3)\}$.

For each configuration, we fit the model using $F \in \{1, 2, 3, 4\}$ NN-GP layers and summarize performance using AUC, MSPE, 95% PI coverage and length over over three simulation replications.

```
# NN-GP Data Generation Simulation Values used in each of the three cases
```

```
# Number of Graph Layers
```

```
L = 2
```

```
# Number of nodes in the graphs on each layer
```

```
J = 20
```

```
# Number of attributes for each node
```

```
m = 2
```

```
# Number of time points (with one being out-of-sample)
```

```
T_len = 21
```

```
# Hyperparameters for NN-GP Data Generation in Cases 1,2, and 3
```

```
# True dimensions of latent vectors used for data generation, R*
```

```
R_vals = c(4, 3, 2)
```

```
# True number of NN-GP layers used for data generation, F*
```

```
sim_F_vals = c(1, 2, 3)
```

```
# Concatenate the corresponding true R and F values into a list of vectors
```

```
R_and_sim_F_vals = I(list(c(R_vals[1], sim_F_vals[1]),  
                           c(R_vals[2], sim_F_vals[2]),  
                           c(R_vals[3], sim_F_vals[3])))
```

```
# Values for F, the number of NN-GP layers, used to fit DJL in each case
```

```
# of the sensitivity analysis
```

```
fitted_F_vals = c(1:4)
```

```
# Random seeds for the replications of each case
```

```
seed_vals = c(22, 25, 26)
```

```
# Create a grid using the true R and F for each case, the value of F used to fit DJL,  
# and the values for the seeds in each replication
```

```
param_grid = expand.grid(R_and_sim_F_vals = R_and_sim_F_vals,
                        fitted_F = fitted_F_vals,
                        seed = seed_vals)
```

```
# Number of MCMC iterations used to fit DJL
niter = 3000
```

```
# Number of iterations to discard as burn-in after fitting DJL
n_burn_in = 2000
```

```
# Dimension of the latent vectors used for fitting DJL
fitted_R = 4
```

Run the Simulations

The following code runs the different simulations in parallel and saves the results.

```
# Create a directory named "outputs" to store results if it does not already exist
if (!dir.exists("outputs")) {
  dir.create("outputs")
}
```

```
# Configure parallel R environment
```

```
# Number of cores to use, if left equal to NULL, all available cores will be used
n_workers = 6
```

```
if (is.null(n_workers)) {
  n_workers = detectCores()
}
```

```
print(paste("Number of cores: ", n_workers, sep = ""))
```

```
cl <- makeCluster(n_workers)
registerDoParallel(cl)
```

```
parallel_runs <- foreach (case = 1:nrow(param_grid)) %dopar% {
```

```
  source("model_generation_running_and_testing_functions.R")
```

```
  cur_params = param_grid[case, ]
```

```
  model_generation_running_and_testing(L = L, J = J, m = m, T_len = T_len,
                                       R = cur_params$R_and_sim_F_vals[[1]][1],
```

```

time_grid = seq(from = 1, to = T_len, length.out = T_len),
time_props_missing = rep(0.1, L),
edge_props_missing = rep(0.25, L), nngp_gen = TRUE,
tergm_gen = FALSE,
nngp_L = cur_params$fitted_F,
sim_nngp_L = cur_params$R_and_sim_F_vals[[1]][2],
sigma2_w = 0.4,
sigma2_b = 0.01,
niter = niter, fitted_R = fitted_R,
fitted_R_zeta = fitted_R,
seed = cur_params$seed,
n_burn_in = n_burn_in, n_thinning = 1,
remove_last_point = TRUE)
}

stopCluster(cl)

```

Aggregate the Results

The section of the code aggregates the results from the simulation replications for $F \in \{1, 2, 3, 4\}$ in Case 1, Case 2, and Case 3, respectively.

```

# Aggregated results for Case 1 with R* = 4, F* = 1
results_case_1 = F_sensitivity_results(fitted_F_vals = fitted_F_vals,
fitted_R = fitted_R,
seed_vals = seed_vals,
L = L, J = J, m = m,
R = R_vals[1],
sim_F = sim_F_vals[1],
T_len = T_len, niter = niter, n_burn_in = n_burn_in)

# Add F heading
results_case_1 <- lapply(results_case_1, function(df) {
df <- as.data.frame(df)
df <- rownames_to_column(df, var = "F")
df
})

# Aggregated results for Case 2 with R* = 3, F* = 2
results_case_2 = F_sensitivity_results(fitted_F_vals = fitted_F_vals,
fitted_R = fitted_R,
seed_vals = seed_vals,
L = L, J = J, m = m,
R = R_vals[2],
sim_F = sim_F_vals[2],
T_len = T_len, niter = niter, n_burn_in = n_burn_in)

# Add F heading
results_case_2 <- lapply(results_case_2, function(df) {
df <- as.data.frame(df)

```

```

df <- rownames_to_column(df, var = "F")
df
})

# Aggregated results for Case 3 with  $R^* = 2$ ,  $F^* = 3$ 
results_case_3 = F_sensitivity_results(fitted_F_vals = fitted_F_vals,
                                       fitted_R = fitted_R,
                                       seed_vals = seed_vals,
                                       L = L, J = J, m = m,
                                       R = R_vals[3],
                                       sim_F = sim_F_vals[3],
                                       T_len = T_len, niter = niter, n_burn_in = n_burn_in)

# Add F heading
results_case_3 <- lapply(results_case_3, function(df) {
  df <- as.data.frame(df)
  df <- rownames_to_column(df, var = "F")
  df
})

```

Simulation Results

The following tables summarize the results for sensitivity to F in Case 1, Case 2, and Case 3, respectively.

Table 1: AUC, MSPE, and 95% Prediction Interval Coverages and Lengths for different values of F used to fit DJL in the in-sample, missing, and out-of-sample predictive scenarios in Case 1. Here, we use $L = 2$, $J = 20$, $T = 20$, $m = 2$, $R^* = 4$, and $F^* = 1$ for data generation. DJL is fitted with $R = 4$. Note. These results correspond to Table 8 in the supplementary materials.

F	1	2	3	4
in	0.9822	0.9649	0.9409	0.9415
mis	0.9891	0.9864	0.9811	0.9844
out	0.9981	0.9980	0.9951	0.9974

(a) AUC for Edge Prediction

F	1	2	3	4
in	0.2870	0.2883	0.3415	0.5741
mis	0.2954	0.2927	0.3471	0.5499
out	0.2831	0.3014	0.3808	0.6305

(b) MSPE for Nodal Attribute Prediction

F	1	2	3	4
in	1.0000	0.9838	0.9671	0.9560
mis	1.0000	0.9750	0.9417	0.9458
out	0.9917	0.9000	0.6583	0.6667

(c) 95% Prediction Interval Coverage

F	1	2	3	4
in	3.7482	2.1035	2.2223	3.4283
mis	3.9641	2.1071	2.2238	3.4235
out	5.7746	2.4053	2.2930	3.4561

(d) 95% Prediction Interval Length

Table 2: AUC, MSPE, and 95% Prediction Interval Coverages and Lengths for different values of F used to fit DJL in the in-sample, missing, and out-of-sample predictive scenarios in Case 2. Here, we use $L = 2$, $J = 20$, $T = 20$, $m = 2$, $R^* = 3$, and $F^* = 2$ for data generation. DJL is fitted with $R = 4$. Note. These results correspond to Table 9 in the supplementary materials.

F	1	2	3	4
in	0.9244	0.9011	0.8659	0.8631
mis	0.9155	0.9084	0.8901	0.8844
out	0.9928	0.9853	0.9826	0.9850

(a) AUC for Edge Prediction

F	1	2	3	4
in	0.3042	0.2810	0.4233	0.5811
mis	0.3152	0.2834	0.4018	0.5266
out	0.2949	0.2922	0.4844	0.6333

(b) MSPE for Nodal Attribute Prediction

F	1	2	3	4
in	0.9866	0.9801	0.9593	0.9569
mis	0.9958	0.9750	0.9292	0.9292
out	0.9417	0.8917	0.6500	0.6667

(c) 95% Prediction Interval Coverage

F	1	2	3	4
in	2.6458	2.4136	2.8971	3.4686
mis	2.7697	2.4463	2.9068	3.4673
out	3.5683	2.7643	2.9529	3.4491

(d) 95% Prediction Interval Length

Table 3: AUC, MSPE, and 95% Prediction Interval Coverages and Lengths for different values of F used to fit DJL in the in-sample, missing, and out-of-sample predictive scenarios in Case 3. Here, we use $L = 2$, $J = 20$, $T = 20$, $m = 2$, $R^* = 2$, and $F^* = 3$ for data generation. DJL is fitted with $R = 4$. Note. These results correspond to Table 10 in the supplementary materials.

F	1	2	3	4
in	0.7460	0.7243	0.6987	0.6959
mis	0.6714	0.6863	0.6657	0.6703
out	0.8828	0.8792	0.8639	0.8622

(a) AUC for Edge Prediction

F	1	2	3	4
in	0.5660	0.5668	0.7311	0.7663
mis	0.5772	0.6259	0.7755	0.8105
out	0.3617	0.3589	0.6523	0.7112

(b) MSPE for Nodal Attribute Prediction

F	1	2	3	4
in	0.9884	0.9806	0.9657	0.9630
mis	0.9833	0.9792	0.9583	0.9542
out	0.9833	0.9417	0.7833	0.7583

(c) 95% Prediction Interval Coverage

F	1	2	3	4
in	3.8890	3.5408	3.7512	3.8374
mis	3.9475	3.5656	3.7532	3.8245
out	5.0399	3.8755	3.7613	3.8382

(d) 95% Prediction Interval Length